

Engineering Notes

Propeller Ground and Ceiling Effect Parametric Data

Jielong Cai* and Sidaard Gunasekaran†
University of Dayton, Dayton, Ohio 45469

Michael OL‡

Folderol, LLC, Xenia, Ohio 45385

and

Anwar Ahmed§

Auburn University, Auburn, Alabama 36849

<https://doi.org/10.2514/1.C035839>

Nomenclature

A	=	propeller disk area, m^2
C_P	=	propellerpower coefficient ($C_P = 2\pi C_Q = P/\rho n^3 D^5$)
C_T	=	propeller thrust coefficient ($C_T = T/\rho n^2 D^4$)
D	=	propeller diameter, m
h	=	distance to ground plane, m
n	=	propeller rotational speed, revolutions per second
P	=	power, W
P_c	=	corrected power, W
P_{IGE}	=	power in-ground effect, W
P_{OGE}	=	power out-of-ground effect, W
Q	=	torque, N·m
R	=	radius, m
S	=	propeller blade projected area, m^2
T	=	thrust, N
γ	=	propeller pitch, in.
ρ	=	air density, kg/m^3
σ	=	propeller solidity

I. Introduction

GROUND effect on rotor/propeller performance has been studied theoretically [1] and experimentally [2–9] for decades. Placing the propulsor adjacent and parallel to a ground-plane offers two benefits: an increase in thrust produced and a decrease in power required for a given rotational speed. Here we consider the ratio of stand-off distance from the ground-plane (h) to propeller diameter (D) down to $h/D = 0.05$, and we attempt to resolve real or perceived inconsistencies in the literature, owing to variations in propeller diameter, pitch, and solidity, which sometimes confound the arrival at a clear trend. We focus on airplane-type (highly twisted) propellers.

Presented as Paper 2019-1097 at the AIAA Scitech 2019 Forum, San Diego, CA, January 7–11, 2019; received 16 December 2019; revision received 7 September 2020; accepted for publication 29 September 2020; published online 23 October 2020. Copyright © 2020 by the American Institute of Aeronautics and Astronautics, Inc. All rights reserved. All requests for copying and permission to reprint should be submitted to CCC at www.copyright.com; employ the eISSN 1533-3868 to initiate your request. See also AIAA Rights and Permissions www.aiaa.org/randp.

*Graduate Student, Mechanical and Aerospace Department. Member AIAA.

†Assistant Professor, Mechanical and Aerospace Department. Member AIAA.

‡Associate Fellow AIAA.

§Professor, Aerospace Engineering. Associate Fellow AIAA.

Propeller ground effect theory dates back at least to Betz's model, in 1937 [1], with later models by Cheeseman and Bennet [6] and Hayden [9]. Figure 1 [2–9] gives clear evidence that in-ground-effect (IGE) thrust will be higher than out-of-ground-effect (OGE; $h/D \rightarrow \infty$), and, correspondingly, the power required will be lower. However, scatter among results and models is considerable and suggests that geometric and aerodynamic variables besides h/D matter. Especially with the result from Knight and Hefner [5], the IGE power is seen to be affected by blade pitch. Blade solidity, or thrust-weighted solidity, is also a factor [10].

There is also the question of the extent to which the direction of thrust matters, that is, whether the slip stream is into, or away from, the ground plane. The latter application is of particular relevance to small-scale unmanned air vehicles. Finally, it would be appealing to find a collapse in the data for various propeller diameters, solidities, and pitch values, onto one curve of P_{IGE}/P_{OGE} versus h/D .

II. Results

A. Experimental Setup

Experiments were conducted on a thrust stand, built from $1.5'' \times 1.5''$ aluminum extrusion bars, at the University of Dayton. All propellers were from the “thin electric” APC series (www.apcprop.com [11]), selected because of evidence in the literature for their relatively high efficiency [12]. All propellers were driven by an E-Flite Power 60–400 kV brushless out-runner electric motor, powered by a PSW 30–108 constant-voltage power supply, and attached to an ATI Industrial Automation Mini-40 six-component force balance (www.ati-ia.com [13]). The data sampling rate was 100 Hz, for a 15 s duration at each data point, with two runs to improve repeatability for each data point (value of propeller rotational speed/diameter/pitch/type, and h/D). To minimize sensor drift, tares were taken before and after each test, and the average was subtracted from the respective thrust and torque readings. To achieve negative h/D (ceiling effect), the propeller was attached reversely on the motor, with the motor rotation direction also reversed. Propeller distance to the ground, h , was measured to the midpoint of the propeller hub.

Propeller parameters are shown in Table 1, with manufacturer-supplied diameter and pitch given in inches. Blade sectional Reynolds number and effective angle of attack are calculated at 70% semispan, for the peak rotational speed achieved for each propeller. The angle of attack is from the blade-element-momentum method of [12]. Propellers labeled “C” in Table 1 are derived from physically cutting off the outboard portion of the blade, to achieve a higher solidity while maintaining pitch. The propeller rotational speed range, with lower bound from signal-to-noise considerations and upper bound limited by the electrical capacity of the power supply, was 3000–6000, depending on propeller diameter.

B. Parameter Studies

A common representation of IGE thrust and torque is “normalized power” (P_{IGE}/P_{OGE}) at constant thrust. To calculate this, IGE and OGE thrust are measured over a range of propeller rotational speeds. A reference value of rotational speed is fixed, at which OGE thrust is noted. For every h/D of interest, one interpolates across the rotational speed values of measured thrust at that h/D , to find that particular rotational speed, for which the thrust equals the aforementioned OGE thrust. Then, at this particular rotational speed, the IGE power is obtained from interpolating across the power measurement data set at that h/D . Results are averaged across OGE thrust levels, that is, across propeller rotational speed values.

Results for P_{IGE}/P_{OGE} are given in Fig. 2, for various pitch-to-diameter ratios (γ/D), with error bars at 95% confidence intervals. P_{IGE}/P_{OGE} does not quite attain 1.0 at either $h/D = 1.5$ or -1.5 .

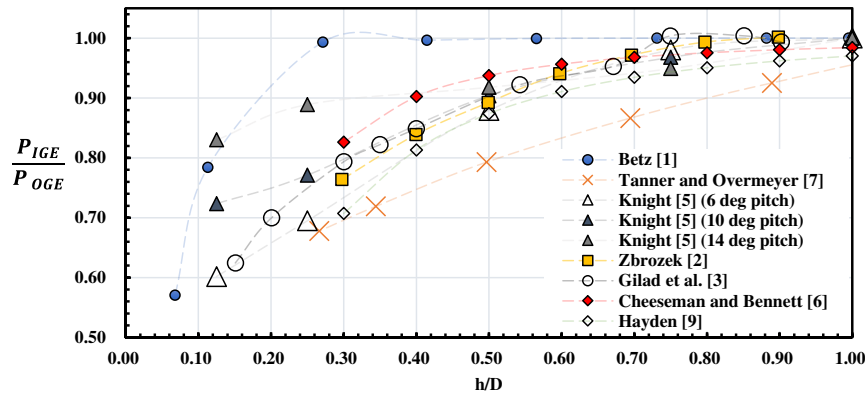


Fig. 1 Ratio of in-ground-effect (IGE) to out-of-ground-effect (OGE) power required at constant thrust vs h/D .

Table 1 Propeller properties

Propeller, diameter $\times$ pitch	Pitch (γ)/diameter (D)	Solidity (σ)	Peak tip speed, m/s	Peak Reynolds number ($\times 1000$)	Weighted thrust coefficient (C_T)/solidity (σ)	Effective angle of attack, deg
17×12	0.705	0.0718	124.6	145	1.401	11.7
17×10	0.588	0.0753	132.1	160	1.378	8.9
17×7	0.412	0.0846	142.0	190	0.889	5.2
$14 \times 7C$	0.500	0.1080	119.9	185	1.123	6.7
14×14	1.000	0.0659	110.6	105	1.623	18.5
14×8.5	0.607	0.0744	117.9	108	1.327	9.3
14×7	0.500	0.0772	122.3	110	1.226	7.0
14×6	0.429	0.0808	122.9	109	0.938	5.5
$11 \times 7C$	0.636	0.1359	96.3	155	1.299	9.2
11×12	1.091	0.0793	94.3	91	1.746	20.2
11×7	0.636	0.0849	100.1	91	1.249	9.6
11×5.5	0.500	0.0896	102.1	92	1.078	6.7
$11 \times 4.5P$	0.409	0.0967	101.8	89	0.754	4.8

Evidently, this is a blockage effect from the thrust stand, unrelated to ground effect. In the ground effect configuration, the thrust stand is upstream of the propeller, whereas in ceiling effect, it is in the propeller wake. The difference of 5% in P_{IGE}/P_{OGE} can thus be associated with installation blockage. This is a fixed bias error. Noting this, we focus on $|h/D| < 1.0$.

For the 17×7 propeller, normalized power at the most extreme ground standoff, $h/D = 0.05$, is half of the OGE value. The trend of P_{IGE}/P_{OGE} versus h/D is symmetric with respect to sign of h/D , but there is a slight asymmetry at $h/D = +1$ versus $h/D = -1$. For the 17×10 propeller, IGE power reduction is much less dramatic. For $h/D \sim -0.2$, there is actually a reversal in P_{IGE}/P_{OGE} . The 11×5.5

and 14×7 propellers, with $\gamma/D = 0.5$, nearly overlap, and except for the smallest h/D , have similar trend to that of the 17×7 . Here 11×7 is an intermediate case. The trend is a strengthening of ground effect with smaller γ/D .

Normalized power ratio P_{IGE}/P_{OGE} potentially masks separate trends of how thrust is increasing or power required is decreasing. Direct presentation of thrust and power coefficients is given in Fig. 3. Trends of Fig. 2 repeat in both subplots of Fig. 3, with the exception that positive-negative h/D symmetry for the 17×7 case is not as evident. Instead, complementary trends in thrust and power combine to produce the symmetry for the 17×7 case in Fig. 2. For both the 17×7 and 11×5.5 propellers, peak thrust increase appears

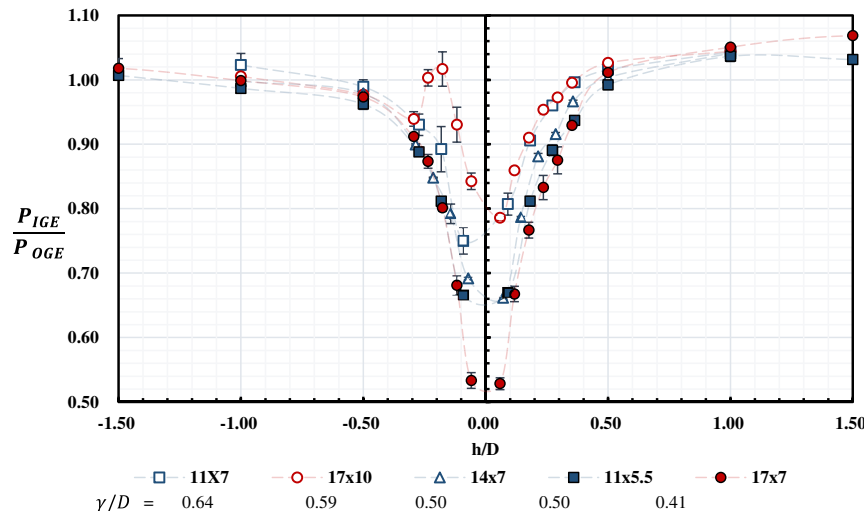


Fig. 2 Normalized power required for constant thrust, for a selection of propellers.

to be at slightly negative h/D , whereas peak power reduction is at slightly positive h/D . The 17×10 case still contains a reversal in both thrust and power at $h/D \sim -0.2$.

We next consider propeller solidity σ . Figure 4 shows P/P_{OGE} versus h/D , for $7''$ pitch propellers: off-of-the-shelf 11×7 and 17×7 , and cut-diameter versions of the $17''$ propeller, resulting in 14×7 or 11×7 diameter: $14 \times 7C$ and $11 \times 7C$, respectively. The 11×7 and $11 \times 7C$ share γ/D (square symbols) as do the 14×7 and $14 \times 7C$

(triangular symbols). For each such pair, the case with higher solidity has deeper P_{IGE}/P_{OGE} trough as $h/D \rightarrow 0$. Perhaps as mere coincidence, the $11 \times 7C$ and 14×7 cases look to be nearly antisymmetric, despite their large disparity in σ (0.1359 and 0.0772, respectively).

C. Synthesis

We propose an empirical fit for collapsing the IGE data to one curve:

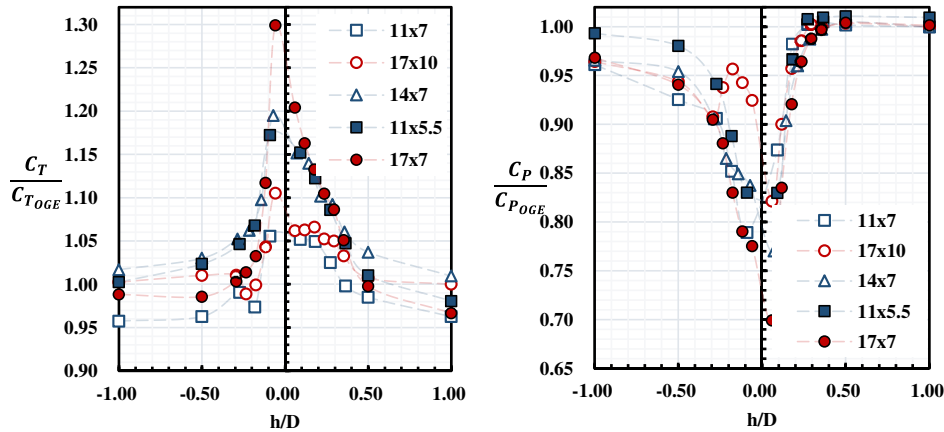


Fig. 3 Ratio of IGE to OGE thrust coefficient (left) and power coefficient (right) for cases of Fig. 2.

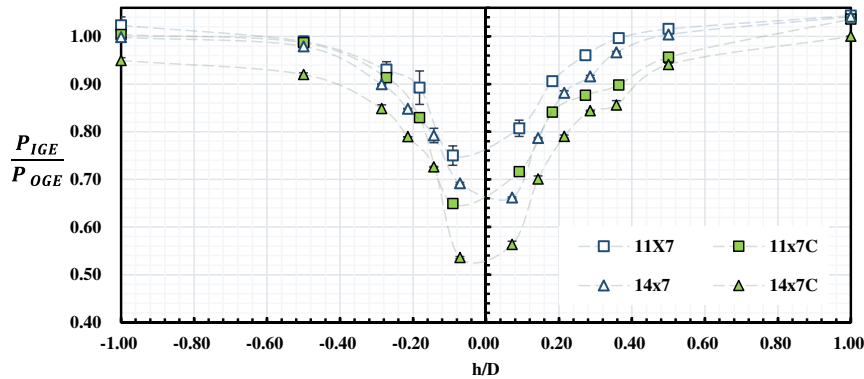


Fig. 4 Normalized power required for constant thrust, varying solidity.

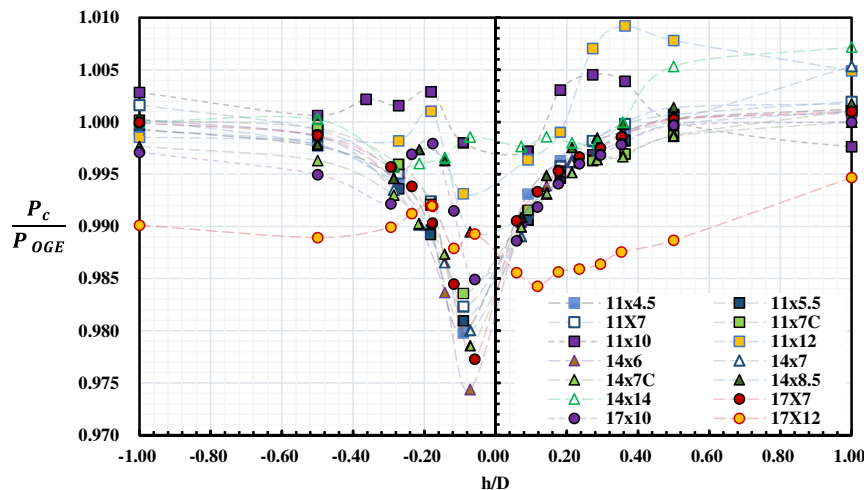


Fig. 5 Empirical fit for normalized power required at constant thrust.

$$\frac{P_c}{P_{\text{OGE}}} = \begin{cases} 1 - \frac{1 - \frac{P_{\text{IGE}}}{P_{\text{OGE}}}}{\sigma * 100} \left(\frac{\gamma}{D}\right)^2 = -0.0513 \left(\frac{h}{D}\right)^2 + 0.0508 \left(\frac{h}{D}\right) + 0.9803 \left(0 \leq \frac{h}{D} \leq 0.5\right) \\ 1 - \frac{1 - \frac{P_{\text{IGE}}}{P_{\text{OGE}}}}{\sigma * 100} \left(\frac{\gamma}{D}\right)^1 = -0.1461 \left(\frac{h}{D}\right)^2 - 0.1218 \left(\frac{h}{D}\right) + 0.9707 \left(-0.5 \leq \frac{h}{D} \leq 0\right) \end{cases} \quad (1)$$

Figure 5 shows the power required at constant thrust, scaled using Eq. (1), for all cases in Table 1. Outlying cases are the 11×10 , 11×12 , 17×12 , 17×10 , and 14×14 , which comparatively have higher γ/D of close to 0.6 or greater; 14×8.5 is partially successful, whereas the other cases evince reasonable collapse to one curve. One might speculate that poor IGE results of higher γ/D propellers, and failure of these cases to collapse to one curve in Fig. 5, are due to blade stall. That is, if ground effect increases the effective angle of attack along the propeller blade, and this angle of attack is already high (Table 1, right-most column), it may exacerbate stall.

III. Conclusions

Ground effect was found, not surprisingly, to increase effective thrust and to decrease power required. Higher solidity and especially lower pitch-to-diameter ratio lead to stronger ground effect. For the highest ratio of pitch to diameter, there are anomalous results in extreme ground effect, possibly due to blade stall. Empirical fit leads to a curve of normalized power at constant thrust, versus ratio of ground standoff to propeller diameter, which collapses fairly well, for all propellers, save for those of the largest pitch-to-diameter ratio. The relation holds for ground effect and ceiling effect, that is, whether the pressure side of the propeller is away from or toward the ground, respectively.

References

- [1] Betz, A., "The Ground Effect on Lifting Propellers," NACA TM 836, 1937.
- [2] Zbrozek, J., "Ground Effect on the Lifting Rotor," British Aeronautical Research Council TR 2347, London, U.K., 1947.
- [3] Gilad, M., Chopra, I., and Rand, O., "Performance Evaluation of a Flexible Rotor in Extreme Ground Effect," *37th European Rotorcraft Forum*, European Rotorcraft Forum, Milan, Italy, Sept. 2011, pp. 816–825.
- [4] Fradenburgh, E. A., "The Helicopter as a Ground Effect Machine," *Journal of the American Helicopter Society*, Vol. 5, No. 4, Oct. 1960, pp. 24–33.
<https://doi.org/10.4050/JAHS.5.4.24>
- [5] Knight, M., and Hefner, R. A., "Analysis of Ground Effect on the Lifting Airscrew," NACA TN-835, Dec. 1941.
- [6] Cheeseman, C., and Bennett, W. E., "The Effect of the Ground on a Helicopter Rotor in Forward Flight," British Aeronautical Research Council TR 3021, London, U.K., 1957.
- [7] Tanner, P. E., and Overmeyer, A. D., "Experimental Investigation of Rotorcraft Outwash in Ground Effect," NASA Langley Research Center NF1676L-21054, May 2015.
- [8] Law, H. Y. H., "Two Methods of Prediction of Hovering Performance," U.S. Army Aviation System Command TR-72-4, St. Louis, MO, Feb. 1972.
- [9] Hayden, J. S., "Effect of the Ground on Helicopter Hovering Power Required," *American Helicopter Society 32nd Annual Forum*, American Helicopter Soc., Washington, D.C., May 1976.
- [10] Leishman, J. G., "Ground Effect," *Principles of Helicopter Aerodynamics*, 2nd ed., Cambridge Univ. Press, Cambridge, U.K., 2006, pp. 257–263.
- [11] APC Thin Electric Propeller, <https://www.apcprop.com/> [retrieved Jan. 2017].
- [12] Ol, M. V., Zeune, C., and Logan, M., "Analytical/Experimental Comparison for Small Electric Unmanned Air Vehicle Propellers," AIAA Paper 2008-7345, Aug. 2008.
<https://doi.org/10.2514/6.2008-7345>
- [13] ATI Industrial Automation, F/T Mini-40 Sensor, <http://www.ati-ia.com/> [retrieved Jan. 2017].